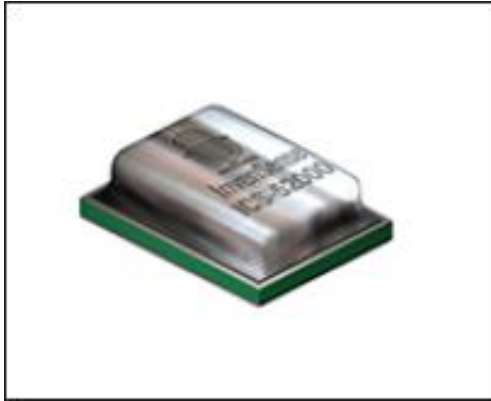


Patronus Acoustics Team

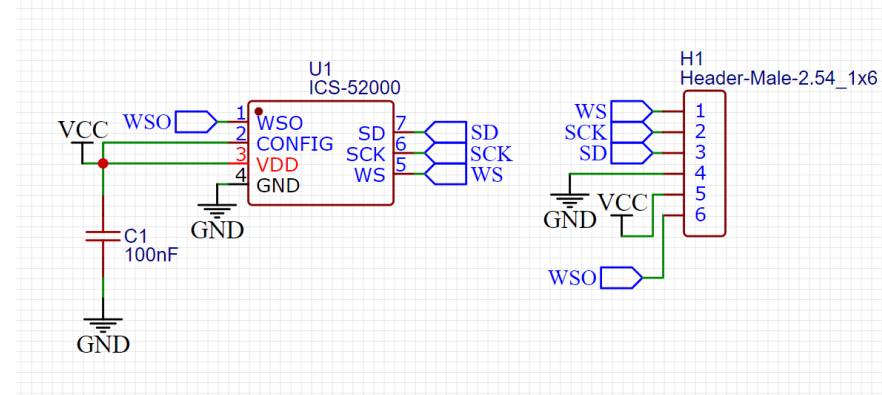
Passive UVA Detection System

Components

ICS-52000 with Custom Breakout Board



+

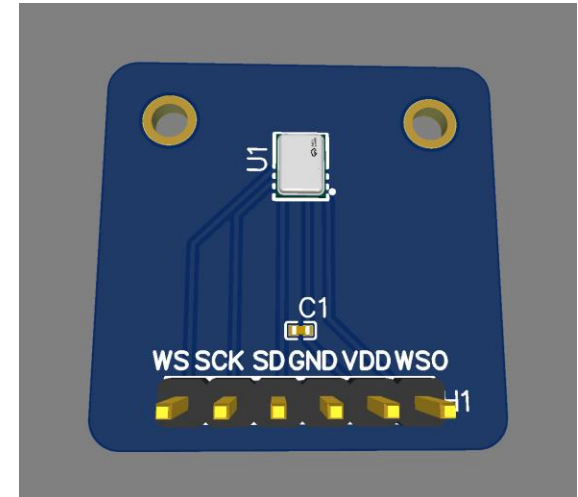
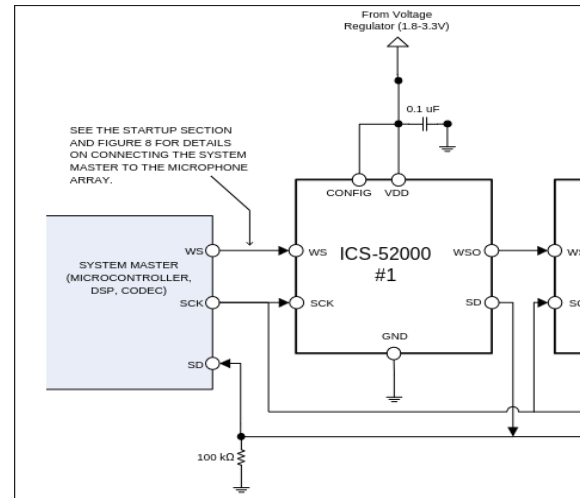


- Datasheet

- <https://d17t6iyxenbwp1.cloudfront.net/s3fs-public/2026-04/ds-000121-ics-52000-datasheet.pdf?VersionId=33j2Nn7def20QHljhHElw1ftv9JV43VN>

Why ICS-52000 with Custom Breakout Board

- ICS-52000 is built specifically for microphone arrays
 - Mic share same clock and data line (put in serial)
- Good quality (signal to noise ratio)
- Custom PCB provides flexibility in mic-array geometry
- Downsides:
 - EOL coming up soon
 - Startup cost



How to buy custom PCB

- Buy from JCLPCB
 - <https://jlcpcb.com/>
- Step 1 – buy ICS-52000 to assembler (JCLPCB) (many e.g. 50)
- Step 2 – give assembler your current PCB design
- Step 3 – order minimum amount of "assembled" components
- Step 4 – test PCB design when it arrives
- Step 6 – iterate/improve on design
- Step 5 – repeat from step 2 if something is wrong!
 - Assembler will hold onto your ~50 ICS parts while you iterate on your design
- (<https://jlcpcb.com/help/article/how-to-consign-parts-to-jlcpcb>)

NUCLEO-H753ZI

- Product
 - <https://no.mouser.com/ProductDetail/STMicroelectronics/NUCLEO-H753ZI?qs=%252B6g0mu59x7JptTWmsgHt6Q%3D%3D>
 - Local retailer: <https://no.farnell.com/stmicroelectronics/nucleo-h753zi/development-board-32bit-arm-cortex/dp/4225705>
- Datasheet
 - Must download pdf from site



Why NUCLEO-H753ZI (or H723ZG)

- SAI peripherals with 16-slot TDM mode
 - Can interface with ICS 52000 easily
- Others have used this MCU for similar purposes
 - <https://jeffmcbride.net/acoustic-beamforming/>
- Comes with Ethernet onboard
- Cheap dev-board
- Cons:
 - Memory limitations unknown

GNSS/GPS module L1 NEO-7M 56ch SMA

- Product
 - <https://www.electrokit.com/en/gps-modul-u-blox-neo-7m-for-arduino>
- Datasheet
 - https://www.electrokit.com/upload/quick/b9/5a/97b0_41022728_Manual.pdf



Why GNSS/GPS module L1 NEO-7M 56ch SMA

- Provides PPS output
- Satisfies sub-ms sync requirements (between station sync)
- Fast prototyping – no custom PCB
- Can easily add antenna if needed

Design

Suggested Design (per stations)

$$d \leq \frac{\lambda}{2} = \frac{v_{\text{sound}}}{2 \times f_{\text{max}}}$$

Our station-build uses two concentric microphone array circles – separated along the boresight axis. The microphone in use is the ICS-52000, which is built specifically for microphone arrays and solves the synchronization problem by sharing the same clock and data line across all microphones using a TDM solution. This geometry allows us to capture a sound sources' azimuth, elevation, and front vs. back trajectory (from the stations perspective).

The outer circle has a diameter of 34 cm and the inner 17 cm. Based on the spatial aliasing formula, the outer array captures sound frequencies up to 500 Hz and the inner up to 1 kHz before aliasing occurs. Lower frequencies are deliberately targeted as they travel farther, directly supporting longer range drone detection.

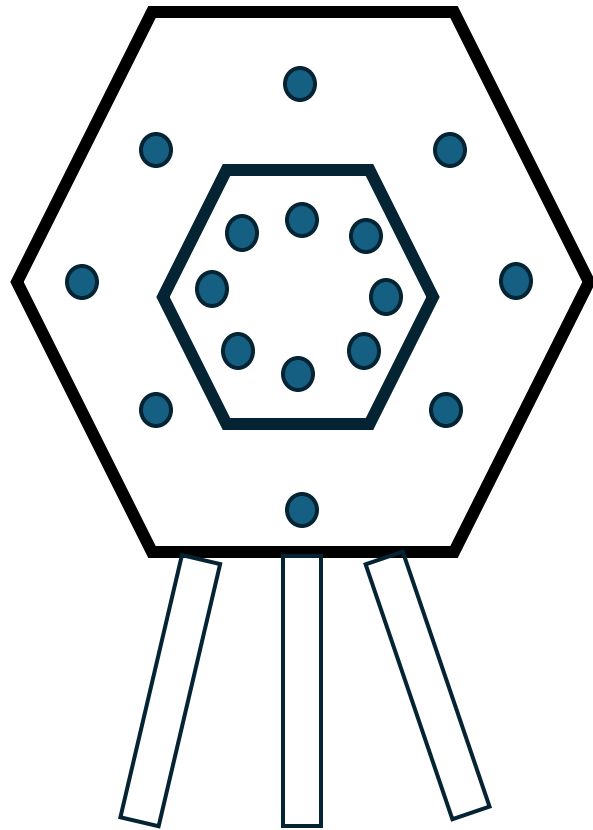
SRP-PHAT – uses beamforming and time difference of arrival (TDOA) over a relevant grid – to enhance the signal and localize the source. The beam becomes progressively tighter and more accurate as more microphones contribute. The captured and enhanced signal is then sent to a C2 system over Ethernet in log-mel format, where a CNN model performs classification.

A single station can resolve only a bearing – an azimuth and elevation pair pointing from the array toward the source. It cannot determine the range on its own. Recovering a 3D position therefore requires at least two stations. Each station's bearing defines a ray in world coordinates, and the closest point of approach between them is computed and taken as the estimated source position. Cross-station time alignment is provided by a dedicated GPS module at each station, with the PPS output as a common reference giving sub-millisecond timestamp alignment between nodes. When matching bearings arrive from both stations within the same time window, C2 can compute the 3D position (and plot it on a map).

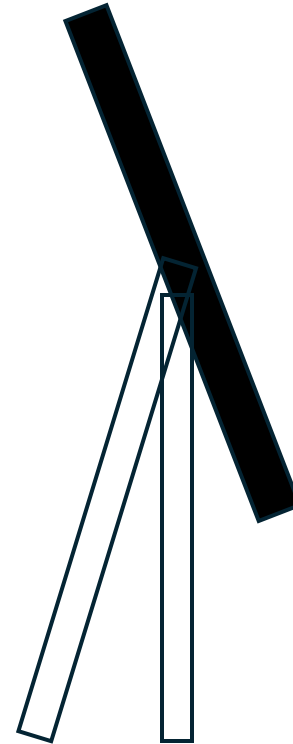
Note: This design is not set in stone and will evolve as development progresses.

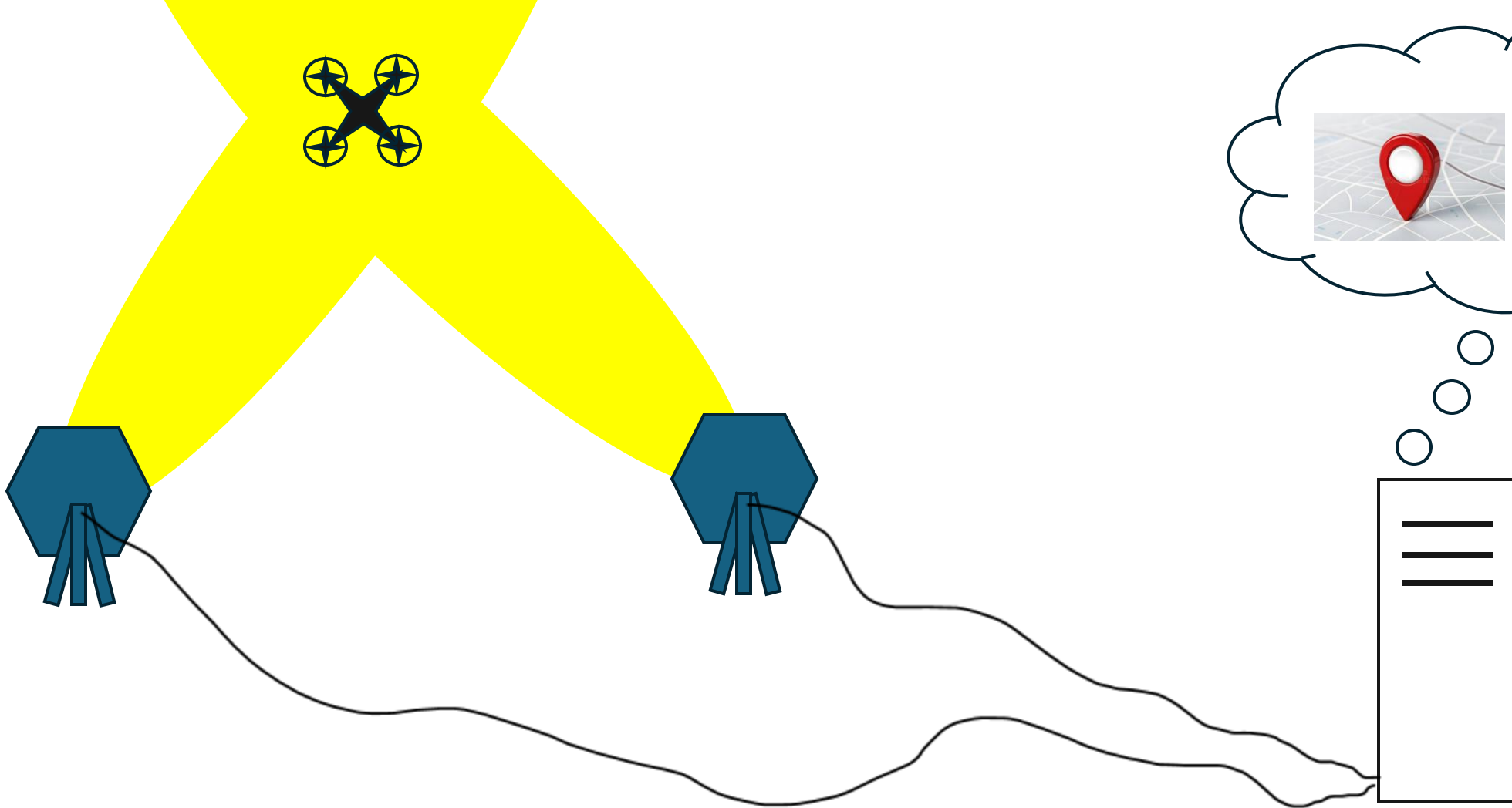
Planned improvements: Real-time drone tracking and simultaneous detection of multiple drones.

Front view

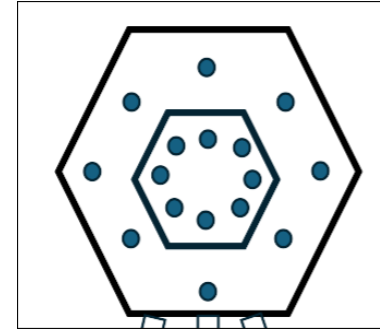


Side view





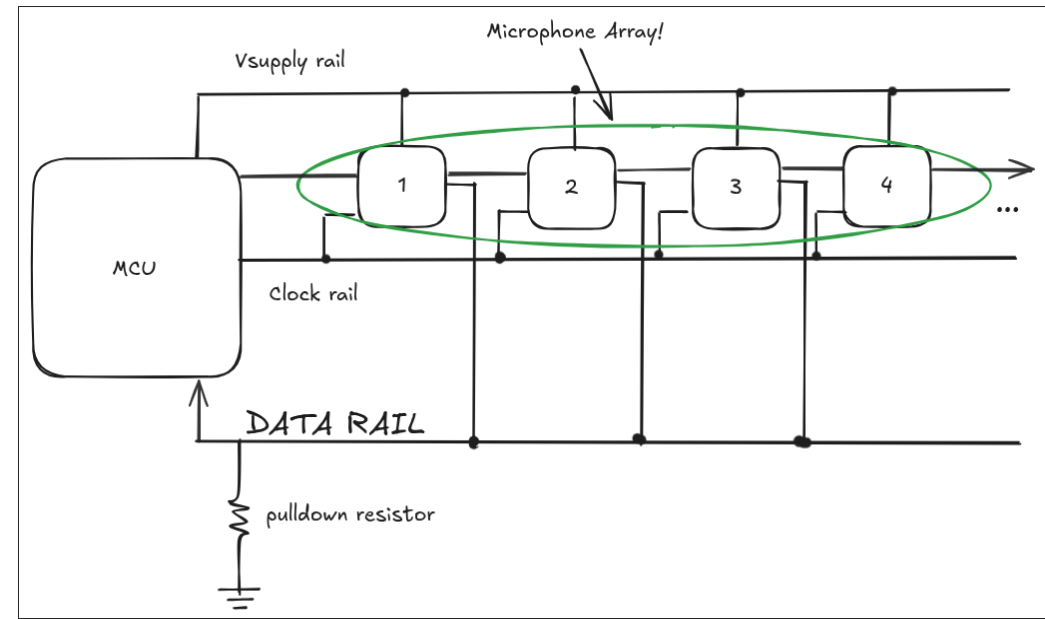
Why this geometric shape?



Two UCA microphone arrays are separated along the boresight axis. The circular geometry of each ring provides azimuth directionality **(5)**. Axially separating the two layers introduces a depth aperture, which lets the array distinguish sources in front of it from sources behind – breaking the mirror symmetry that a single planar ring cannot resolve on its own. Beamforming over the combined geometry will be used to enhance the steered signal and suppress off-axis noise **(7)**. Optimal spacing for the two rings is not yet fixed, but simulation tools and code will be used to characterize the trade-off space before fabrication.

Why the ICS-52000

These microphones have built in TDM making it much easier to synchronize clocks. Microphones in the same array will share data – and clock line. The ICS-52000 has a high SNR compared to other microphones in the same price range (65 dB). Similar SNR ratio mic used in this project **(2)**. Using PDM microphones requires filtering to PCM samples which is computationally heavy **(1)**.



Why log-mel and CNN?

Acoustic UAV detection is heavily challenged by environmental noise. The traditional detection approach utilizes thresholding on extracted features (using FFT), but is often outperformed by newer methods. The newer methods either combine traditional and machine learning techniques or goes full out on ML. Popular implementations use MLP, Random Forest and SVM **(3)** - however, newer studies point to CNN (and RNN) as the better solution – in combination with log-mel formatted audio data **(4)**. Hybrid architectures, combining classical spectral filtering with ML have shown great promise for robust performance in real-world conditions **(3)**.

We decided to go hybrid as it fits well with our goals and budget. Each station runs a classical detection (and beamforming) algorithm while a stronger (on-hand) C2 system does CNN classification on the log-mel datastream it receives from the station. This separation is made to avoid overspending on microcontrollers and to delegate station-to-C2 responsibilities cleanly.

Expectations – bring-up of 8-16 mics (one station)

- Mic station:
 - Azimuth and elevation estimate (SRP-PHAT algorithm)
 - Detection range ca. 10 – 50m (hopefully closer to upper range)
 - Stream log-mel features continuously to C2
 - Stream PPS/time sampled DOA (az. and el.) to C2
- C2:
 - CNN classification on log-mel features

Expectations – full prototype (two stations)

- Mic stations:
 - Azimuth and elevation estimate (SRP-PHAT algorithm)
 - Detection range ca. 20 – 70m (hopefully closer to upper range)
 - Stream log-mel features continuously to C2
 - Stream PPS/time sampled DOA (az. and el.) to C2
- C2:
 - CNN classification on log-mel features
 - Triangulation of DOA received from stations
 - Estimate a 3D location of drone

NB: This is far into the future!

Early Iteration Resources

- <https://github.com/jewhoguy/SRP-PHAT> (simulate SRP-PHAT)
- https://github.com/kbhujbal/SudarshanChakra-acoustic_uav_threat_detection_CNN (log-mel + CNN)
- <https://github.com/SOUNDS-RESEARCH/DroneSynthesis2DOA> (more log-mel + CNN)
- <https://github.com/mcbridejc/acoustic-array-tools> (STM32 + beamforming implementation)

References

- (1) <https://jeffmcbride.net/acoustic-beamforming/>
- (2) <https://medium.com/data-science/audio-deep-learning-made-simple-sound-classification-step-by-step-cebc936bbe5>
- (3) <https://ui.adsabs.harvard.edu/abs/2026AcAcu..10...12G/abstract>
- (4) <https://ieeexplore.ieee.org/document/7829341>
- (5) <https://www.qualcomm.com/news/onq/2021/02/fundamentals-voice-ui-part-ii-designing-optimum-beamformers?utm>
- (6) <https://link.springer.com/article/10.1007/s42417-022-00562-4>
- (7) <https://sistc.com/blog-microphone-beamforming-fundamentals/>